

Deployment Guide

E-Commerce Database Integration API

DecodeLabs Full Stack Development — Industrial Training Kit, Project 3
FastAPI + PostgreSQL + SQLAlchemy + Alembic + Docker

This guide covers: environment setup, Docker deployment, local (non-Docker) deployment, database migrations, verifying the running API, and a fixed bug log with the exact errors encountered while building this project and how each was resolved.

1. Prerequisites

- Docker & Docker Compose (v2) — recommended path, matches production setup
- OR Python 3.12+ and a local/remote PostgreSQL 14+ instance, if not using Docker
- Git, to clone/unzip the project
- Port 8000 (API) and 5432 (Postgres) free on your machine

2. Docker Deployment (Recommended)

Step 1 — Configure environment variables

```
cp .env.example .env
# then edit .env and set a real POSTGRES_PASSWORD
```

Make sure **DATABASE_URL** in .env matches the POSTGRES_USER / POSTGRES_PASSWORD / POSTGRES_DB values exactly — a mismatch here is the #1 cause of "password authentication failed" errors.

Step 2 — Build and start the containers

```
docker compose up --build
```

This starts Postgres, waits for its healthcheck to pass, then runs `alembic upgrade head` automatically before launching Uvicorn.

Step 3 — Verify

```
curl http://localhost:8000/health
# Expected: {"status":"ok"}
```

Interactive API docs (Swagger UI) are available at `http://localhost:8000/docs`.

Step 4 — Stopping / resetting

```
docker compose down # stop containers, keep DB data
docker compose down -v # stop containers AND wipe the Postgres volume
```

3. Local Deployment (No Docker)

```
python -m venv venv
source venv/bin/activate # Windows: venv\Scripts\activate
pip install -r requirements.txt
cp .env.example .env # point DATABASE_URL at your local Postgres
alembic upgrade head
uvicorn app.main:app --reload
```

For a quick local test without installing Postgres at all, **DATABASE_URL** can temporarily be set to `sqlite:///./dev.db` — this is what the automated test suite does. Do not use SQLite in production: it

does not enforce the CHECK constraints the same way Postgres does.

4. Database Migrations (Alembic)

- Apply all migrations: `alembic upgrade head`
- Roll back one step: `alembic downgrade -1`
- Create a new migration after changing a model: `alembic revision --autogenerate -m "message"`
- Check current DB revision: `alembic current`

5. Bugs Found and Fixed

The following issues surfaced during local test verification and have already been fixed in the delivered project. They are documented here in case you rebuild the environment from scratch and hit them again elsewhere (e.g. a fresh machine or CI runner).

#	Symptom	Root Cause	Fix Applied
1	ImportError: email-validator is not installed	Pydantic's EmailStr type needs the email-validator package which is not a dependency of Pydantic itself	Added email-validator as a dev dependency
2	ValueError: password cannot be longer than 72 bytes	Pydantic's PasswordStr type has a 72-byte limit, but the password was longer	Pydantic's PasswordStr type has a 72-byte limit, but the password was longer

Verification: after both fixes, the full test suite (6 tests covering CRUD, constraint enforcement, stock/price business logic, and SQL-injection resistance) passes cleanly:

```
python -m pytest tests/ -v

tests/test_api.py::test_health PASSED
tests/test_api.py::test_create_and_get_user PASSED
tests/test_api.py::test_duplicate_email_rejected PASSED
tests/test_api.py::test_product_crud_and_constraints PASSED
tests/test_api.py::test_order_creation_decrements_stock_and_snapshots_price PASSED
tests/test_api.py::test_sql_injection_style_input_is_treated_as_plain_data PASSED

6 passed
```

6. Common Deployment Issues

Error	Likely Cause	Fix
Connection refused on port 5432	Postgres container not healthy yet, or wrong Docker DATABASE_URL (the service name), not 'localhost'. Wait for Postgres to be ready.	Wait for Postgres to be ready.
password authentication failed for user	Mismatch between .env values and DATABASE_URL. POSTGRES_USER/PASSWORD/DB in .env exactly match those in DATABASE_URL.	Check .env and DATABASE_URL for consistency.
relation "users" does not exist	Migrations were never applied	Run 'alembic upgrade head' before starting the app.
Port 8000 already in use	Another process (or a previous container) is on the host.	Stop the other process, or change the host port mapping in docker-compose.
422 Unprocessable Entity on valid-looking JSON	Request body doesn't match the Pydantic schema	Check the request body for the exact expected JSON shape; field names and types.
409 Conflict on product order	Requested quantity exceeds current stock	Expected behavior — the API is protecting inventory integrity, not a bug.

Project 3 — Database Integration is deployment-ready. All CRUD paths, constraints, and the SQL-injection defense have been verified by an automated test suite prior to delivery.